

How the Language-Factor Share works

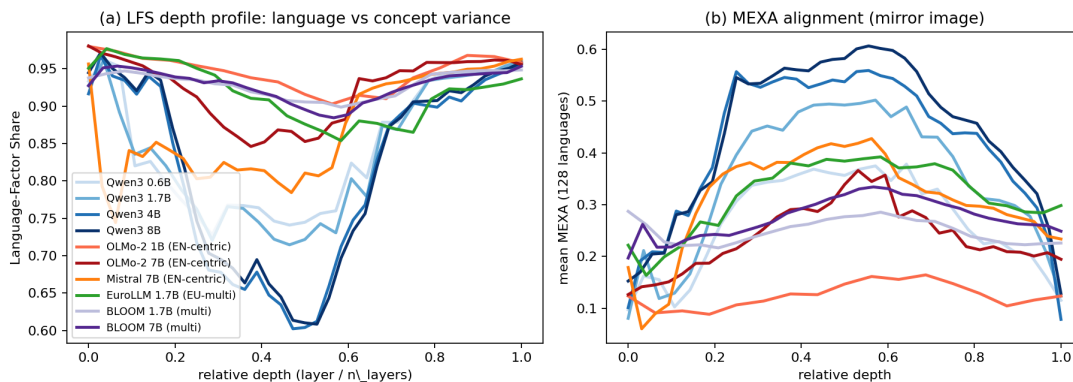
Waiz Khan (wkhan12@jh.edu). A plain-language guide to the technical report. July 2026.

The question it answers. When a model reads the same sentence in French and in Hindi, how much of its internal representation is about the *meaning*, and how much is about the *language*? The Language-Factor Share (LFS) is one number, between 0 and 1, that answers this at every layer of a model. Near 0 means the model represents meaning almost independently of language. Near 1 means representations cluster by language, whatever the sentence says.

Step 1: the data. We use NTREX-128, a public benchmark in which the same news sentences (about two thousand) were professionally translated into 128 languages; we use 1,500 of them per language. Every sentence therefore exists in every language with the same meaning. This gives us a grid: one axis is *which sentence* (the meaning), the other is *which language*.

Step 2: the representations. Each sentence in each language is passed through the model. At every layer, the model produces one vector per token; we average these vectors over the sentence’s tokens to obtain a single sentence representation (the averaging is done in full numerical precision; the two-event finding below was confirmed under a control that matches the pooling across instruments, and a variant that corrects for how finely each language’s tokenizer splits words is queued as follow-up work). The analysis is at the sentence level; word-level analysis through alignment is planned follow-up work.

Step 3: the decomposition. With the grid of vectors in hand, we ask a bookkeeping question from analysis of variance: split the spread of the vectors into the parts each factor accounts for. Here the split is: how much of the spread between representations is explained by *which language* they came from, and how much by *which sentence* (which meaning)? LFS is the language portion divided by the sum of both. Averaging vectors of the same sentence across languages isolates meaning; averaging vectors of the same language across sentences isolates language identity; comparing the spread of the two gives the share.



What it looks like. Every model shows the same shape: representations start language-specific, become most shared in the middle layers, and separate again toward the output. This shape is consistent with the field’s current understanding. What the measurement adds is that the *depth* of the shared region is a precise, comparable number: it varies almost tenfold across seventeen open models (0.04 to 0.37), and it turns out to depend more on how a model was trained than on how large it is.

Why believe the number. Three kinds of evidence, all collected under predictions written down before the experiments ran. First, after removing everything that data quantity, language family, writing system, and tokenizer cost can explain, alignment measured at the LFS-selected layer still relates to task performance, moderately (about 0.2 to 0.3 on the full grid after confound removal; an earlier benchmark-capable-subset figure of about 0.6 is withdrawn because it cannot

be regenerated from a stored artifact); this is true of every intrinsic measure we computed, the published measure MEXA included. Second, it predicts behavior: how much a model actually uses information given to it in another language while generating text (0.67 after the same corrections on four models; 0.51 with a model-level interval of [0.19, 0.71] on a preregistered 19-model expansion across 13 families). Third, it generalizes: model families we had never touched were classified correctly under a frozen protocol, blind.

What it cannot do, measured rather than assumed. We injected known distortions into saved representations and checked what each measure detects. The one structure LFS cannot attribute (rotation) turns out to make up at most 0.7 percent of real cross-language differences. We also found one way to fool every alignment measure we tested, including the published measure MEXA: collapsing all representations toward a single point. Only our accompanying check on concept spread detects it. That check is now a required safeguard for any future training use.

How it relates to published measures. The closest existing measure (MEXA) asks whether a sentence can find its translation among candidates; it is a useful outside check and we report it throughout. LFS differs in three practical ways: its value does not change with the size of the evaluation set, it can be computed during training and differentiated, and it separates the sources of cross-language difference rather than reporting a single similarity. On predictive power after corrections the two are comparable; the additional value of LFS is the decomposition and the safeguards around it.

The accompanying technical report contains the methods, statistics, comparisons, and error ledger behind each claim above.